

Chapter no. 03

Carbohydrates

• Define:-

Carbohydrates are the organic compound made up of carbon, hydrogen and oxygen, typically in a ratio 1:2:1.

They are one of the primary macronutrients alongside proteins and fats, and serve several essential functions in the body.

• Types:-

• Simple carbohydrates:-

These are the sugars like glucose, fructose that provide quick energy. They are found in fruits, milk and sweeteners.

• Complex carbohydrates:-

These include starches and fibers, which are made up of longer chains of sugar molecules. They are found in foods like grains, legumes.

and vegetables.

Functions:-

- * Energy Source:- Carbohydrates are a primary energy source of the body, especially for the brain and muscles for later use.
- * Storage:- Excess carbohydrates can be stored as glycogen in the liver and muscles for later use.
- * Digestive health:- Dietary fiber, a type of carbohydrate, aids digestion and helps maintain gut health.

Role of each Element in Carbohydr-

Carbon:-

Forms the backbone of carbohydrate molecules. The number of carbon atoms can vary, leading to different types of carbohydrates (e.g., glucose has six carbon atom).

Hydrogen:-

Combines with carbon and oxygen to create the structure of carbohydrates and is essential for energy storage and release.

Oxygen:-

Present in the hydroxyl (-OH) groups that define sugars and play a role in energy metabolism.

Structure of Carbohydrates

Monosaccharides:- Simple sugar like glucose) consist of single sugar unit.

Disaccharides:- (Like sucrose) are formed from two monosaccharides.

Polysaccharides:- (like starch and cellulose) are long chains of monosaccharides units.

Polysaccharides:-

Polysaccharides are complex carbohydrates composed of long chains of monosaccharides unit linked together by glycosidic bonds. They serve various functions, including energy storage and structural support in living organisms.

Types:-

1- Storage Polysaccharides:-

Starch:-

Found in plants, Starch is a major energy storage polysaccharides

- composed of amylose (linear chains) and amylopectin (branched chains)
- Easily digestible by human and animals.

Glycogen:-

The primary energy storage polysaccharide in animals, mainly found in the liver and muscles.

- Highly branched structure, allowing for rapid mobilization of glucose when energy is needed.

2- Structural Polysaccharides:-

* Cellulose

- A key component of plant cell walls, providing rigidity and strength.
- Composed of long, unbranched chains of β -glucose units, linked by β -1,4-glycosidic bonds.
- Humans lack the enzymes to digest cellulose, making it a significant source of dietary fiber.

* Chitin:-

- Found in the exoskeletons of arthropods and the cell walls of fungi.
- Composed of N-acetylglucosamine units, forming strong.

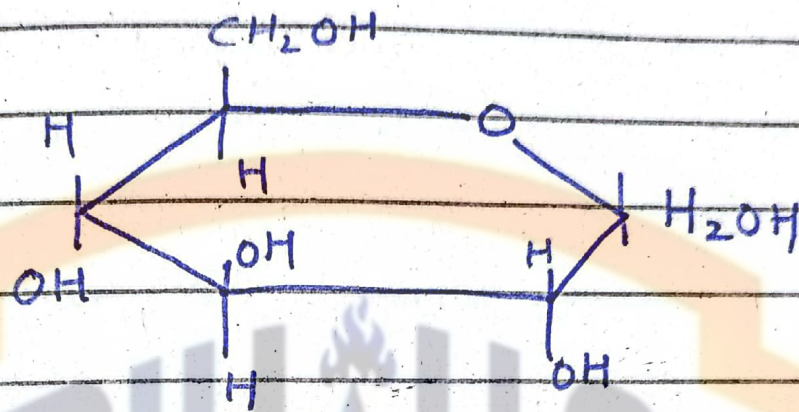
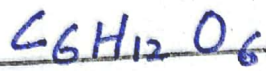
Photosynthesis

Photosynthesis is the biochemical process by which green plants, algae, and some bacteria convert high energy, usually from the sun, into chemical energy in the form of glucose. The process primarily occurs in the chloroplast of plant cells, where chlorophyll captures light energy.

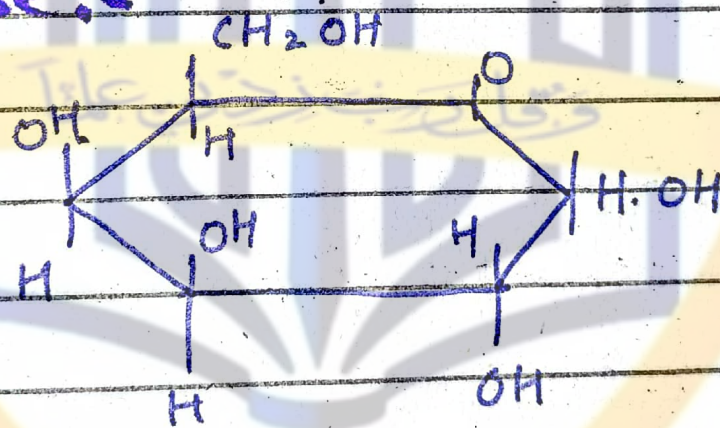
Sugar

Sugar refers to a type of carbohydrates that is sweet, soluble in water and used by living organisms for energy. It typically comes in two forms.

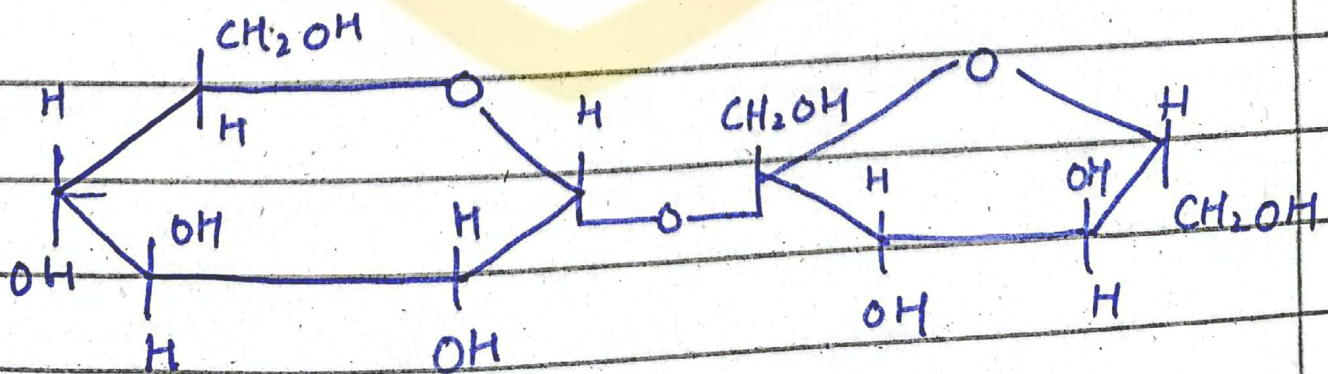
Glucose is a simple sugar or monosaccharide, that serve as a primary energy source for the body cell.



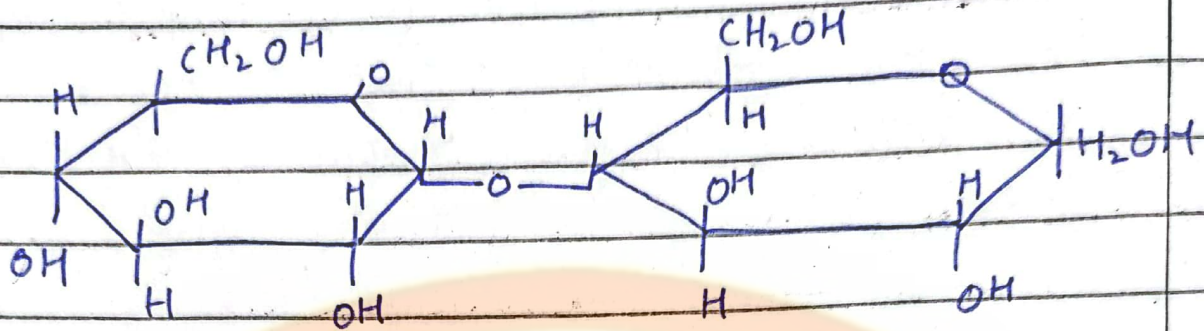
Glactose:-



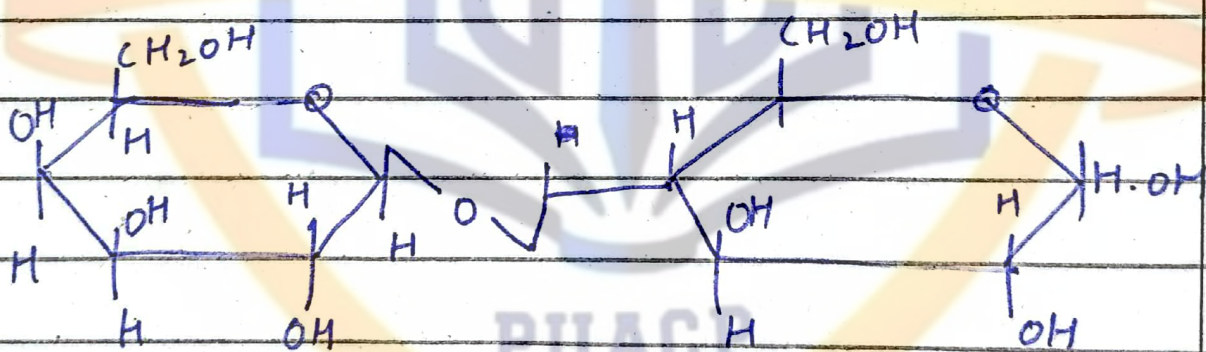
Sucrose



Maltose:-



Lactose:-



Cellulose

Cellulose is a complex carbohydrate or polysaccharide, that is composed of long chains of glucose molecules. It is the main structural component of plant cell wall providing "rigidity" and "strength" to plants. In nature it is one of the most abundant organic material as it is present in the cell wall of many types of plants like algae, and some bacteria.

* Functions of cellulose:-

- 1- Structural Support in plants:- cellulose provide rigidity and strength to the plant cell walls, helping plants to maintain their shapes and resist external pressure. It allows plants to stand upright and grow tall without the need for bone or other internal ^{structure}.
- 2- Protection:- By forming a tough layer around a plant cells, cellulose helps protect from physical damage, pathogens and dehydration.
- 3- Water Regulation:- cellulose combined with other molecule in the cell walls,

helps regulate the flow of water into and out of cells which is essential for maintaining cellular pressure (turgor pressure and overall plant health).

4- **Dietary Fiber:-** Although indigestible to humans, cellulose adds bulk to the diet as fiber. It aids digestion by promoting regular bowel movements, improving gut health, and potentially lowering risks of certain diseases.

* **Properties:-** Polysaccharides composition:- composed of beta-D-glucose units linked by $\beta(1-4)$ glycosidic bonds, cellulose has a stable and repetitive structure that contributes to its resistance to enzymatic degradation by most organisms, except those with specialized cellulose enzymes.

* **Use:-** The modified properties of CMC make it ideal for application in food, pharmaceuticals, and cosmetics as a thickener, stabilizer, etc. and use as a gum.

* **Pectins:-** Pectins are a group of complex polysaccharides found naturally in the cell walls of plants, particularly in fruits.

1- **composition:-** These are composed of rhamnogalacturonans (D-galacturonic acid and rhamnose) act as "cement" that helps bind plant cells together.

* **Characteristics:-**

* **Gelling ability:-** Pectins can form gels when mixed with sugar and acid or with calcium ions. This gelling property is a defining characteristic and is a crucial product like jams, jellies & fruit preservatives.

* **Polysaccharides composition:-** Pectins are complex polysaccharides linked by (1-4 glycosidic bonds). They also contain natural sugars like rhamnose, arabinose and galactose.

* **Water solubility:-** These are water soluble especially when they are broken down into smaller molecules or when modified their hydrophilic groups such as carboxyl ($-\text{COOH}$) and hydroxyl ($-\text{OH}$) groups attract water and allow them to

* Functions of Pectins:-

• Structural role in Plants:-

- 1- **Cell wall strength**:- Pectins are major components of the plant cell wall, where they help bind cells together, providing structural support and rigidity to the plant tissue.
- 2- **Maintaining plant cells integrity**:- They contribute to the integrity of plant cells by forming a gel like matrix that helps maintain the shape and structure of plant tissue.

• Water Retention:-

- 1- **Water regulation**:- Pectin helps to hold water within the plant cell wall contributing turgidity and hydration. This is crucial for maintaining cell pressure and preventing wilting.

Gums

Gums are a group of complex carbohydrates polysaccharides that are typically derived from plant or microbial sources.

• Functions of Gum

1- Stabilizer:-

- ⊙ Polysaccharides gums stabilize emulsions, suspensions and foams by preventing phase separation
- ⊙ Example:- Xanthan gum in salad dressing and icecreams
gums Arabic in beverages.

2- Water Retention:-

Gums are excellent at water binding water due to their hydrophillic nature, which helps maintain moisture in products.

Mucilages

Mucilages is a thick, gelatinous substance composed primarily of polysaccharides, produced by plants, algae and certain microorganisms. It is hydrophillic, meaning it readily absorb water to form a gel-like consistency.

* Functions of Mucilages:-

- 1- Water storage:- Helps plant retain moisture
- 2- Seed germination:- Mucilages around seeds absorb water and aids in sprouting.
- 3- Protection:- Shields plants from dehydration pathogens and medical damages.

Agar

Agar is a natural gelatinous substance derived from the cell walls of certain species of red algae, primarily Gelidium and Gracilaria. It is mainly composed of two polysaccharides, agarose and agarpectin.

* Functions of Agar:-

- 1- Food industry:- used in soap Jell, ice creams, meat and pastes etc.